

Date	31 January 2025	Maximum Marks	4 Marks
Team ID	SWTID-2026-5019	Project Name	Import Data using Transform Maps (Spreadsheet)

DESIGN THINKING • IDEATION PHASE

Empathize & Discover

ServiceNow System Administration
ETL & Data Integration Track

PROJECT OVERVIEW & BUSINESS CONTEXT

This micro project focuses on importing structured data from an external spreadsheet into the ServiceNow platform using **Import Sets** and **Transform Maps**. The project simulates a real-world enterprise scenario where bulk employee data is received in Excel format and must be migrated into ServiceNow efficiently, safely, and accurately.

The spreadsheet data is first loaded into an **Import Set table** (serving as an isolated staging buffer), and a **Transform Map** is configured to map source Excel columns to target table fields (such as sys_user). Upon executing the transformation, records are created or updated, and verified against schema constraints to ensure total data integrity.

Why Empathy Mapping in Data Migration?

Data migration is frequently treated as a purely mechanical task, yet user error, schema mismatches, and duplicated identities have profound human impact. Empathy mapping uncovers the administrator's fears, procedural pressures, and cognitive load when dealing with uncleaned spreadsheets.

Target Persona & Operational Goal

Persona: Junior ServiceNow Administrator / HR Systems Specialist.

Goal: Seamlessly transform hundreds of legacy employee rows into active user profiles with 100% field parity and zero unintended duplicate accounts.

Empathy Map Canvas Architecture (6-Dimension Analysis)

1. THINK & FEEL

- Inner worries about corrupting production database tables.
- Concerns over ambiguous dates, choices, and coalesce keys.

2. SEE

- Dirty Excel sheets with missing headers or irregular formats.
- Staging table rows (u_imp_*) vs. destination live records.

3. HEAR

- Urgent management requests to onboard new staff immediately.
- Senior architects warning against uncoalesced transforms.

4. SAY & DO

- Executing "Auto Map Matching Fields" and manual field scripts.
- Running trial transforms and auditing Import Log error rows.

Pains (Fears, Frustrations & Bottlenecks)

Overwritten records, duplicate employee profiles, unparsed choice lists, and manual entry fatigue.

Gains (Wants, Needs & Success Measures)

Automated repeatable import pipelines, verified data integrity, zero manual typos, and fast turnaround.

Empathy Map Canvas: ServiceNow Data Migration

Case: Bulk Employee Spreadsheet Import using Import Sets & Transform Maps

Ideation • Completed Canvas

What does the Administrator THINK AND FEEL?		Worries, internal aspirations, schema hesitations, and preoccupations	
FEAR OF DATA CORRUPTION "What if bad data in this Excel file accidentally overwrites active employee records in production?"	COALESCE ANXIETY "Did I select the right unique key (User ID / Email)? If coalesce is false, it will flood the system with duplicates."		
RELIEF IN STAGING ISOLATION "Import Sets give me peace of mind—the raw data stays isolated in the staging table until I approve the transform."	FIELD FORMATTING DOUBTS "Will date formats (DD/MM vs MM/DD) or choice values like 'Department' fail during target field mapping?"		

What do they HEAR?	Stakeholders, peers & leadership	 ServiceNow Admin Data Specialist Migrating bulk Excel data to sys_user table	What do they SEE?	Environment, screens & artifacts
HR / MANAGEMENT "We have 250 new personnel in Excel; they must have accounts in ServiceNow before morning."			RAW SPREADSHEETS Excel files with inconsistent headers, mixed capitalization, and unstandardized phone formats.	
SENIOR ARCHITECT "Always test your Transform Map in a sub-production instance and review the transform log first."			SERVICENOW CONSOLE Import Set staging records (u_imp_employee) waiting with status 'Pending Transform'.	
COLLEAGUES "Be careful with blank spreadsheet cells; they can clear existing values if not handled."			TARGET SCHEMA The target table schema (sys_user) with mandatory reference fields and unique keys.	

What do they SAY AND DO?		Observable behaviors, procedural actions & technical practices	
LOAD DATA SOURCE Uploads the spreadsheet via 'Load Data', specifying file path, sheet number, and staging table name.	AUTO & MANUAL MAPPING Uses 'Auto Map Matching Fields' for identical names, then manually links u_emp_id to user_name.		
CONFIGURE COALESCE Explicitly sets Coalesce = true on the Employee ID column to prevent duplicate user creation.	TRANSFORM & VALIDATE Executes the transform run, audits the Transform History log (inserts vs. updates), and spot-checks profiles.		

PAIN		GAIN	
Fears, Frustrations & Obstacles		Wants, Needs & Success Metrics	
DUPLICATE USER ACCOUNTS Transform running without coalesce creates duplicate active profiles, breaking license counts and ticketing.	DATA TYPE MISMATCHES Text strings failing to resolve into Reference fields (e.g. Department, Manager, Location).	100% DATA INTEGRITY Flawless migration where every row matches target schema requirements without data truncation.	REPEATABLE ETL PIPELINE The configured Transform Map can be reused next quarter for recurring bulk spreadsheet imports.
TEDIOUS MANUAL ENTRY		RAPID EXECUTION Bulk migration finishes in under 2 minutes with	